

Níckolas de Aguiar Alves (he/him)

 PhD candidate in physics

 Federal University of ABC

 Portuguese (native), English (fluent), Italian (basic)

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Education

2023–
Sep.



Doctor in Physics

Federal University of ABC (UFABC), Santo André, São Paulo, Brazil

Advisor: André G. S. Landulfo 

2021–2023
Feb. Apr.



Master in Physics

Federal University of ABC (UFABC), Santo André, São Paulo, Brazil

Advisor: André G. S. Landulfo 

Thesis: *Nonperturbative Aspects of Quantum Field Theory in Curved Spacetime*

2017–2020
Feb. Dec.



Bachelor in Physics

Institute of Physics, University of São Paulo (IFUSP), São Paulo, São Paulo, Brazil

Advisor: João C. A. Barata 

Appointments

2026–2027
Aug. Aug.



Visiting PhD Candidate

Scuola Normale Superiore di Pisa (SNS), Pisa, Toscana, Italy

Supervisor: Guilherme L. Pimentel 

Funded by the São Paulo Research Foundation (FAPESP) Grant No. [2026/02509-8](#)

Publications

Research Articles

- 1 N. Aguiar Alves and A. G. S. Landulfo. *Sound as a gauge theory and its infrared triangle*. 2026. arXiv: [2512.15796 \[hep-th\]](#). Submitted.
- 2 N. Aguiar Alves and A. G. S. Landulfo. “Null infinity as a Killing horizon”. *Phys. Rev. D* **112.6**, 065017 (2025). arXiv: [2504.12514 \[gr-qc\]](#).
- 3 N. Aguiar Alves, A. G. S. Landulfo, and B. A. Costa. “Positive mass in general relativity without energy conditions”. *Phys. Rev. D* **111.4**, 044027 (2025). arXiv: [2408.00154 \[gr-qc\]](#).

Review Articles

- 1 N. Aguiar Alves. “Lectures on the Bondi–Metzner–Sachs group and related topics in infrared physics”. *Eur. Phys. J. C* **86.5**, 507 (2026). arXiv: [2504.12521 \[gr-qc\]](#).

Pedagogical Articles

- 1 N. Aguiar Alves. “The unexpected effectiveness of the Picard maneuver: Seeing three images of the same superluminal spaceship”. *Am. J. Phys.* arXiv: [2608.23624 \[physics.pop-ph\]](#). Forthcoming.
- 2 N. Aguiar Alves and B. A. Costa. “Liénard–Wiechert potentials and the electromagnetic memory effect”. *Am. J. Phys.* Forthcoming.
- 3 C. C. Rodrigues Evangelista and N. Aguiar Alves. “Using thermodynamics to learn gravitational wave physics”. *Eur. J. Phys.* **47.2**, 025604 (2026). arXiv: [2602.21261 \[gr-qc\]](#).

Essays and Short Contributions

- 1 N. Aguiar Alves. “Physics as a branch of poetry”. *Am. J. Phys.* **94.7** (2026), p. 507.
- 2 N. Aguiar Alves and B. A. Costa. *The Measure of a Mass*. 2025. arXiv: [2503.18963 \[gr-qc\]](#). Pre-published.

Theses

- 1 N. Aguiar Alves. “Nonperturbative Aspects of Quantum Field Theory in Curved Spacetime”. MSc thesis. Santo André, Brazil: Federal University of ABC, 2023. xxiv, 152 pp. arXiv: [2305.17453 \[gr-qc\]](#).

Other Works

- 1 N. Aguiar Alves. “Quantum Field Theory in Curved Spacetime. An Introduction”. Unpublished lecture notes. 2023.

In Preparation

- 1 N. Aguiar Alves, A. G. S. Landulfo, and A. D. Pereira. *Nonperturbative renormalization group flow for a particle detector*. In preparation.
- 2 N. Aguiar Alves, C. C. Rodrigues Evangelista, G. J. Olmo, and D. Rubiera-Garcia. *Platypus stars: Exotic compact objects supported by vacuum pressure*. In preparation.

Funding

2026–	PhD Scholarship – Research Internship Abroad Explorations with Twistors and Cosmological Correlators São Paulo Research Foundation (FAPESP) Grant No. 2026/02509-8 Advisor in Brazil: André G. S. Landulfo  Supervisor abroad: Guilherme L. Pimentel 
2025–	PhD Scholarship The Sky as a Killing Horizon and Other Topics on the Infrared Structure of Gravity São Paulo Research Foundation (FAPESP) Grant No. 2025/05161-0 Advisor: André G. S. Landulfo 
2023–2025	PhD Scholarship Coordenação de Aperfeiçoamento de Pessoal de Ensino Superior (CAPES) Granted to pursue my PhD after placing first in UFABC’s admission ranking
2021–2023	MSc Scholarship The Functional Renormalization Group in Quantum Field Theory in Curved Spacetimes São Paulo Research Foundation (FAPESP) Grant No. 2021/07372-7 Advisor: André G. S. Landulfo 
2021	MSc Scholarship Coordenação de Aperfeiçoamento de Pessoal de Ensino Superior (CAPES) Granted to pursue my MSc after placing first in UFABC’s admission ranking
2019–2020	BSc Scholarship Hyperbolic Equations São Paulo Research Foundation (FAPESP) Grant No. 2019/12158-4 Advisor: João C. A. Barata 

Academic Service

Conferences Organized

- 2026  [Thermality in Quantum Field Theory in Curved Spacetimes](#)
Main organizer
Promoted by the [ICTP South American Institute for Fundamental Research](#)
- 2023  [Golden Wedding of Black Holes and Thermodynamics](#)
Main organizer

Schools Organized

- 2024–  [São Paulo School in Gravitational Physics](#)
Main organizer
Part of [ICTP Physics Without Frontiers](#) (second edition onward)
- 2018–2021  [Jayme Tiomno School in Theoretical Physics](#)
Founder and organizer of the first three editions

Peer Review

- 2026–  Reviewer for [Physical Review D](#)
-  Reviewer for the [American Journal of Physics](#)

Scientific Societies

- 2025–  [Brazilian Physical Society](#)
- 2024–  [International Society for Quantum Gravity](#)

Teaching Experience

Minicourses and Workshops

- 2025  [Introduction to Black Hole Physics](#)
[II São Paulo School in Gravitational Physics](#)
Minicourse for early undergraduate students
- 2024  [Infrared Symmetries of General Relativity](#)
[I São Paulo School in Gravitational Physics](#)
Minicourse for advanced undergraduate and early graduate students
- 2023  [Quantum Field Theory in Curved Spacetime](#)
[Golden Wedding of Black Holes and Thermodynamics](#)
Minicourse for early graduate students
- 2021  [Algebraic Methods of Theoretical Physics](#)
[III Jayme Tiomno School in Theoretical Physics](#)
Minicourse on group theory in physics for first-year undergraduate students
- 2020  [An Introduction to \$\LaTeX\$](#)
[XV Oceanography Thematic Week](#)
Oceanographic Institute, University of São Paulo
- 2018  [An Introduction to \$\LaTeX\$](#)
[I Jayme Tiomno School in Theoretical Physics](#)

Teaching Assistant

- 2025  [Topics in Celestial Holography](#)
Institute of Physics, University of São Paulo
Graduate-level course

Teaching Experience (continued)

- 2024  Advanced Topics in General Relativity
Institute of Physics, University of São Paulo
Graduate-level course
- 2024  Electromagnetic Phenomena
Federal University of ABC
Undergraduate-level course covering introductory electrodynamics
- 2021  Classical Mechanics II
Federal University of ABC
Undergraduate-level course covering analytical mechanics
- 2019  Mathematical Physics I
Institute of Physics, University of São Paulo
Undergraduate-level course covering introductory Fourier analysis

Participation in Panels

- 2026  [VI ICTP-SAIFR Summer School for Young Physicists](#)
Panel with graduate students in physics

Outreach and Miscellanea

-  [VI ICTP-SAIFR Summer School for Young Physicists](#)
Tutored a course on black hole physics for high school students
- 2025  [International Young Physicists Tournament \(Brazil\)](#)
Juror for the final stage of the 2025 edition of the IYPT Brazil
-  [V ICTP-SAIFR Summer School for Young Physicists](#)
Tutored a course on black hole physics for high school students
- 2018–  [Physics Stack Exchange](#)
Active contributor to the Physics Stack Exchange Q&A website
Most prolific contributor to the [qft-in-curved-spacetime](#) tag

Awards

- 2025  “Outstanding Recognition” award at the [UFABC Physics Week](#)
 - awarded to 12 out of 45 participants based on talks about their research progress

Press Coverage

This press coverage list is not exhaustive, as other media outlets might use a few main ones as sources.

- 1 S. Nogueira. [Físico brasileiro demonstra matematicamente sucesso da Manobra Picard](#). Trek Brasilis, 2026. On **N. Aguiar Alves**. “The unexpected effectiveness of the Picard maneuver: Seeing three images of the same superluminal spaceship”. *Am. J. Phys.* arXiv: [2608.23624](#) [[physics.pop-ph](#)]. Forthcoming.
- 2 S. Nogueira. [Brasileiro demonstra manobra consagrada em “Jornada nas Estrelas”](#). Folha de S. Paulo, 2026. On **N. Aguiar Alves**. “The unexpected effectiveness of the Picard maneuver: Seeing three images of the same superluminal spaceship”. *Am. J. Phys.* arXiv: [2608.23624](#) [[physics.pop-ph](#)]. Forthcoming.
- 3 M. von Hippel. [Physicist does the math on Star Trek’s “Picard maneuver”. The trick from Next Generation would be even more effective in real life](#). Ars Technica, 2026. On **N. Aguiar Alves**. “The unexpected effectiveness of the Picard maneuver: Seeing three images of the same superluminal spaceship”. *Am. J. Phys.* arXiv: [2608.23624](#) [[physics.pop-ph](#)]. Forthcoming.

Research Presentations

Contributed Talks

- 1 *Sound as a gauge theory and its infrared triangle.* **Simons–Balseiro School and Workshop on Advanced Aspects of QFT**, Instituto Balseiro, May 18–27, 2026.
- 2 *Quantum Gravity and the Night Sky.* **UFABC Physics Week**, Federal University of ABC, Oct. 14–17, 2025.
- 3 *Positive mass in general relativity without energy conditions.* **XLVII Paulo Leal Ferreira Congress**, Institute of Theoretical Physics, São Paulo State University, Oct. 29–Nov. 1, 2024.
- 4 *Nonperturbative renormalization group flow for a particle detector.* **Golden Wedding of Black Holes and Thermodynamics**, Online, Dec. 4–8, 2023.

Poster Presentations

- 1 *You can teach gravitational wave physics using thermodynamics.* **Third generation gravitational wave detectors: The view from Latin America**, Principia Institute, June 29–July 3, 2026.
- 2 *We can use the renormalization group to study the nonperturbative behavior of particle detectors.* **Simons–Balseiro School and Workshop on Advanced Aspects of QFT**, Instituto Balseiro, May 18–27, 2026.
- 3 *Linear sound has asymptotic symmetries.* **XI QCD Meets Gravity Conference**, Principia Institute, Dec. 15–19, 2025.
- 4 *Linear sound has asymptotic symmetries.* **XI QCD Meets Gravity School**, Institute of Theoretical Physics, São Paulo State University, Dec. 8–12, 2025.
- 5 *Stars with extreme pressures can mimic black holes.* **XLVIII Paulo Leal Ferreira Congress**, Institute of Theoretical Physics, São Paulo State University, Oct. 14–17, 2025.
- 6 *Killing horizons naturally define asymptotic symmetries.* **First International Latin American Conference on Gravitational Waves**, National Institute for Space Research, Sept. 15–19, 2025.
- 7 *Killing horizons naturally define asymptotic symmetries.* **VI Siembra–HoLAGrav Young Frontiers Meeting at ICTP–SAIFR**, Institute of Theoretical Physics, São Paulo State University, June 30–July 11, 2025.
- 8 *Killing horizons naturally define asymptotic symmetries.* **Program on Anomalies, Topology and Quantum Information in Field Theory and Condensed Matter Physics**, Principia Institute, June 16–27, 2025.
- 9 *Negative masses are unstable and we don't need energy conditions to prove it.* **Witnessing Quantum Aspects of Gravity in a Lab**, Principia Institute, Sept. 23–27, 2024.
- 10 *Nonperturbative aspects of quantum field theory in curved spacetime.* **XLVI Paulo Leal Ferreira Congress**, Institute of Theoretical Physics, São Paulo State University, Oct. 24–27, 2023.
- 11 *Nonperturbative aspects of quantum field theory in curved spacetime.* **V Jayme Tiomno Physics School**, Institute of Physics, University of São Paulo, Sept. 4–8, 2023.
- 12 *Nonperturbative aspects of quantum field theory in curved spacetime.* **Interfaces Between Quantum and Classical Statistical Mechanics**, Institute of Mathematics and Statistics, University of São Paulo, July 24–28, 2023.

Flash Talks

- 1 *Nonperturbative renormalization group flow for a particle detector.* **School on Phase Transitions and Gravitational Waves**, International Institute of Physics, Federal University of Rio Grande do Norte, Mar. 4–8, 2024.
- 2 *Nonperturbative renormalization group flow for a particle detector.* **Quantum Spacetime and the Renormalization Group**, Sant'Elmo Beach Hotel, Oct. 2–6, 2023.

Seminars

- ① *Sound as a gauge theory and its infrared triangle.* [Simons Collaboration on Celestial Holography Zoom Seminars](#), Online, Mar. 10, 2026.
- ② *Null infinity as a Killing horizon.* Center for Relativity and Cosmology, Troy University, Aug. 27, 2025.
- ③ *The sky as a Killing horizon.* [Siembra-HoLAGrav Future of Physics](#), Online, May 8, 2025.

I have also given dozens of group seminars at the Federal University of ABC and at the University of São Paulo, which I omit here for conciseness.